CHAPTER 8

# SEQUENCES AND SERIES

**CHAPTER ESSENCE AT A GLANCE:**

Patterns in numbers! In an AP (Arithmetic Progression), you ADD the same common difference d at each step. In a GP (Geometric Progression), you MULTIPLY by the same common ratio r at each step. Infinite geometric series can add up to a clean finite number!

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Geometric Progression (GP):** Sequence a, ar, $ar^2$, $ar^3$... with first term a and common ratio r = $a_n$ over $a_{n-1}$. nth term: $a_n = a\ r^{n-1}$.
- **Sum of n Terms of GP:** $S_n$ = a (1 - $r^n$) over (1 - r) (for r < 1) or a ($r^n$ - 1) over (r - 1) (for r > 1).
- **Sum to Infinity of GP:** S ∞ = a over (1 - r), strictly valid ONLY when |r| < 1. If |r| ≥ 1, series diverges!
- **Geometric Mean (GM):** Single GM between positive numbers a and b is G = sqrt(ab). For any two positive numbers: AM ≥ GM.

## ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$a_n = a \cdot r^{n-1} \quad \text{(General Term of Geometric Progression)}$$

*Geometric Progression nth Term Formula*

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1} \quad (r \neq 1)$$

*Finite Sum of Geometric Progression*

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

*Sum to Infinity of an Absolutely Convergent GP*

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab} \rightarrow A \geq G \quad (\text{AM} \geq \text{GM})$$

*Arithmetic Mean - Geometric Mean Fundamental Inequality*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Infinite GP formula S ∞ = a over (1 - r) works ONLY when the common ratio magnitude is strictly less than 1 (|r| < 1). If |r| ≥ 1, the series diverges!
- **Alert:** If 3 numbers are in AP, choose them as (a - d, a, a + d) so their sum cancels out d immediately!

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

### 1. Find the sum to infinity of the GP: 1, -1 over 3, 1 over 9, -1 over 27... [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Parameters]** a = 1, common ratio r = (-1 over 3) over 1 = -1 over 3. Since |r| = 1 over 3 < 1, the sum to ∞ exists. [1 Mark]

• **[Step 2: Sum Formula]** S ∞ = a over (1 - r) = 1 over [1 - (-1 over 3)] = 1 over (1 + 1 over 3) = 1 over (4 over 3) = 3 over 4. [1 Mark]

• **[Boxed Result: Sum to ∞ S ∞ = 3 over 4]**

### 2. Insert two numbers between 3 and 81 so that the resulting sequence is a GP. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Setup]** Let the sequence be 3, G1, G2, 81. Here a = 3, and 4th term a4 = 81. [1 Mark]

• **[Step 2: Common Ratio & Means]** a4 = a $r^3$ ⇒ 81 = 3 $r^3$ ⇒ $r^3$ = 27 ⇒ r = 3. G1 = a r = 3 × 3 = 9. G2 = a $r^2$ = 3 × $3^2$ = 27. [1 Mark]

• **[Boxed Result: Two numbers are 9 and 27]**

### 3. The 5th, 8th, and 11th terms of a GP are p, q, and s respectively. Show that $q^2$ = p s. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Express Terms]** Let first term be a and common ratio be r: 5th term p = a $r^4$ ---(1). 8th term q = a $r^7$ ---(2). 11th term s = a $r^{10}$ ---(3). [1 Mark]

• **[Step 2: Evaluate $q^2$]** $q^2$ = $(a\ r^7)^2$ = $a^2$ $r^{14}$. [1 Mark]

• **[Step 3: Evaluate p s]** p x s = (a $r^4$) x (a $r^{10}$) = $a^2$ r^(4 + 10) = $a^2$ $r^{14}$. Since $q^2$ = p s = $a^2$ $r^{14}$, the identity is proved. [1 Mark]

• **[Boxed Result: $q^2$ = p s proved]**

### 4. Find the sum of the first n terms of the GP: 1, 2 over 3, 4 over 9... [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Parameters]** a = 1, r = (2 over 3) over 1 = 2 over 3. Since r < 1: $S_n$ = a (1 - $r^n$) over (1 - r). [1 Mark]

• **[Step 2: Substitution & Evaluation]** $S_n$ = 1 [1 - (2 over 3)$^n$] over [1 - 2 over 3] = [1 - (2 over 3)$^n$] over (1 over 3) = 3 [1 - (2 over 3)$^n$]. [2 Marks]

• **[Boxed Result: $S_n$ = 3 [1 - (2 over 3)$^n$]]**

### 5. The sum of first three terms of a GP is 13 over 12 and their product is -1. Find the common ratio and the three terms. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Choice of Terms]** Let the three terms be a over r, a, a r. Product = (a over r) x a x (a r) = $a^3$ = -1 ⇒ a = -1. [1 Mark]

• **[Step 2: Sum Equation]** a over r + a + a r = 13 over 12 ⇒ -1 over r - 1 - r = 13 over 12 ⇒ -(1 over r + 1 + r) = 13 over 12 ⇒ ($r^2$ + r + 1) over r = -13 over 12 ⇒ 12 $r^2$ + 12 r + 12 = -13 r ⇒ 12 $r^2$ + 25 r + 12 = 0. [1 Mark]

• **[Step 3: Factoring & Terms]** 12 $r^2$ + 16 r + 9 r + 12 = 0 ⇒ 4r(3r + 4) + 3(3r + 4) = 0 ⇒ (3r + 4)(4r + 3) = 0 ⇒ r = -4 over 3 or r = -3 over 4. For r = -4 over 3: terms are 3 over 4, -1, 4 over 3. For r = -3 over 4: terms are 4 over 3, -1, 3 over 4. [1 Mark]

• **[Boxed Result: r = -4 over 3 or -3 over 4; Terms are 4 over 3, -1, 3 over 4]**

### 6. Find the sum to n terms of the series: 5 + 55 + 555 + ... to n terms. [Score: 4 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Factor Out 5]** $S_n$ = 5 [1 + 11 + 111 + ... to n terms] = (5 over 9) [9 + 99 + 999 + ... to n terms]. [1 Mark]

• **[Step 2: Setup as Powers of 10]** = (5 over 9) [(10 - 1) + ($10^2$ - 1) + ... + ($10^n$ - 1)] = (5 over 9) [(10 + $10^2$ + ... + $10^n$) - n]. [1.5 Marks]

• **[Step 3: Final Formula]** = (5 over 9) [ {10 ($10^n$ - 1) over 9} - n ] = 50 over 81 ($10^n$ - 1) - 5n over 9. [1.5 Marks]

• **[Boxed Result: $S_n$ = (50 over 81)($10^n$ - 1) - 5n over 9]**

**7. (a) If the pth, qth and rth terms of a GP are a, b and c respectively, prove that a^(q - r) b^(r - p) c^(p - q) = 1. (3 Marks)n(b) If the A.M. of two positive numbers is 10 and their G.M. is 8, find the numbers. (1 Mark) [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Express Terms]** Let first term be A and common ratio be R: a = A R^(p - 1), b = A R^(q - 1), c = A R^(r - 1). [1 Mark]

• **[Step 2: Multiply and Evaluate Powers]** a^(q - r) b^(r - p) c^(p - q) = [A R^(p - 1)]^(q - r) [A R^(q - 1)]^(r - p) [A R^(r - 1)]^(p - q). Power of A: (q - r) + (r - p) + (p - q) = 0 $\Rightarrow A^0 = 1$. Power of R: (p - 1)(q - r) + (q - 1)(r - p) + (r - 1)(p - q) = (pq - pr - q + r) + (qr - qp - r + p) + (rp - rq - p + q) = 0 $\Rightarrow R^0 = 1$. Product = 1 × 1 = 1. Hence proved. [2 Marks]

• **[Step 3: Numerical AM and GM]** A = 10, G = 8. Numbers = $A \pm \sqrt{(A^2 - G^2)} = 10 \pm \sqrt{(100 - 64)} = 10 \pm \sqrt{(36)} = 10 \pm 6 \Rightarrow$ 16 and 4. [1 Mark]

• **[Boxed Result: (a) Product = 1 proved; (b) Numbers are 16 and 4]**